



RETURNS, GROWTH & DEPRECIATION

MCQ FAST SOLVE GUIDE — BY THUNDERSTUDY



TRICK 1 — The Magic Multiplier

Instead of complex formulas, just use a single multiplier:



Growth (r% increase): Multiplier = $1 + r/100$ (e.g., 10% → **1.10**)



Depreciation (r% decrease): Multiplier = $1 - r/100$ (e.g., 10% → **0.90**)

$$\text{Final Value} = \text{Initial Value} \times (\text{Multiplier})^n$$



TRICK 2 — Power Rule (Time Shortcut)

Don't calculate yearly. Raise the multiplier to the power of n (years).

Example: ₹100 grows at 10% for 3 years: $100 \times (1.1)^3 = 133.1$. Done!



TRICK 3 & 4 — The Rule of 72 (MCQ Weapon)

To find when an amount **DOUBLES** (Growth) or **HALVES** (Depreciation):

$$\text{Years} \approx 72 / \text{Rate}$$

Example: Rate = 12%. Years = $72 / 12 = 6$ years. Instant Answer.

TRICK 5 — Successive Changes Cheat

If a value increases 10% then decreases 10%, it's NOT the same!

$(1.10) \times (0.90) = 0.99$ (Final is 99% of original — always a loss).

Mental Multiplier Table

Change	Multiplier	Change	Multiplier
5% Increase	$\times 1.05$	10% Decrease	$\times 0.90$
10% Increase	$\times 1.10$	20% Decrease	$\times 0.80$
25% Increase	$\times 1.25$	5% Decrease	$\times 0.95$

10-Second MCQ Flow

1. Identify if it is **Growth** or **Depreciation**.
2. Convert % into the **Multiplier**.
3. Raise to the **Power** (years).
4. Multiply once and match the closest option.
5. **Eliminate**: If Growth, remove smaller options; if Depreciation, remove larger ones.

★ MEMORY LINE (Revise Before Exam)

"Plus means grow, minus means slow —
Make multiplier, power it, answer will show."

THUNDERSTUDY: CALCULATE LESS, THINK MORE